AI-powered semantic search

in relational databases

using SQL Server 2025 and Ollama

Local embeddings, vector storage and semantic indexing on WideWorldImporters

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SQL Server 2025 | Ollama | WideWorldImporters

From text search to semantic search

- Relational databases store much of the critical information used by organizations.
- Traditional searches depend on exact text matches.
- With embeddings and vector indexes, we can search by meaning while preserving the relational model.
- Project: semantic search with SQL Server 2025, Ollama and WideWorldImporters.

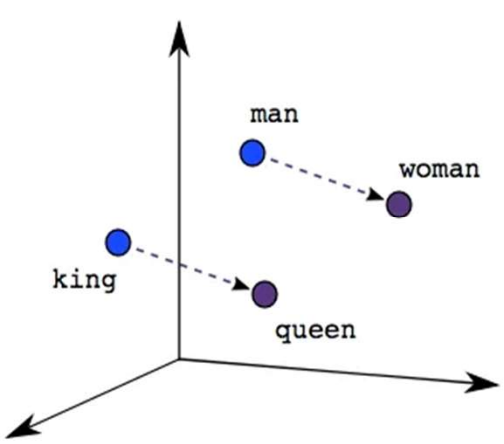
Key concepts: from text to vector

- **Text search: finds word or pattern matches.**
- **Semantic search: searches by meaning, even when words do not match exactly.**
- **Embedding: a numerical vector representation of text, an image or another type of data.**
- **Vector: an ordered list of numbers; in this project, vector(768).**

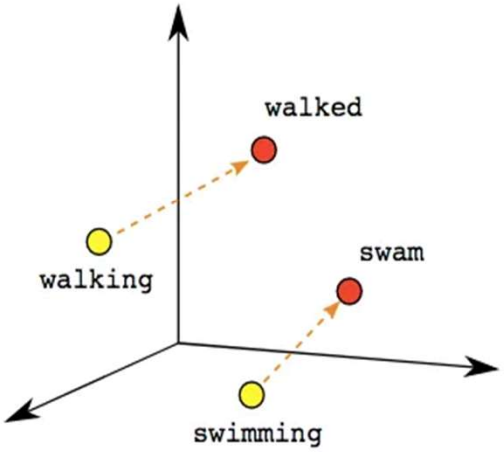
We convert products and queries into vectors using an AI model, then search for the nearest vectors to retrieve the most semantically related products.

Embeddings: vector representations of meaning

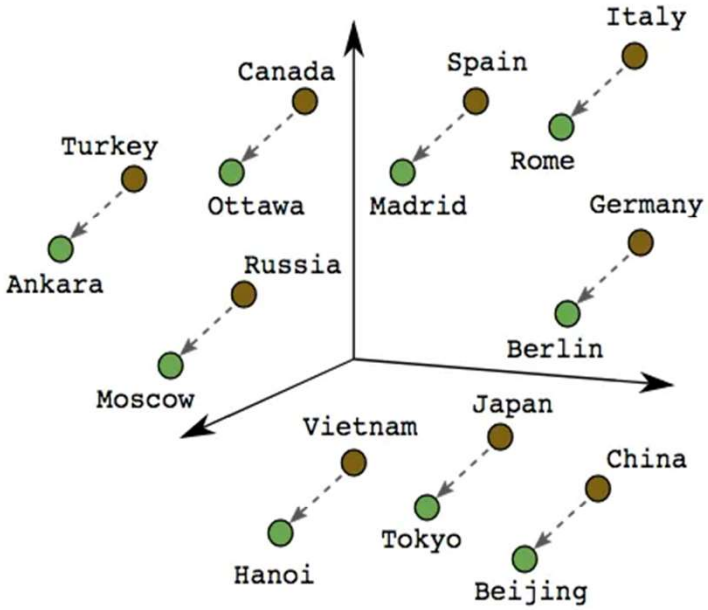
Items with similar meaning tend to be close to each other in vector space.



Male-Female



Verb Tense

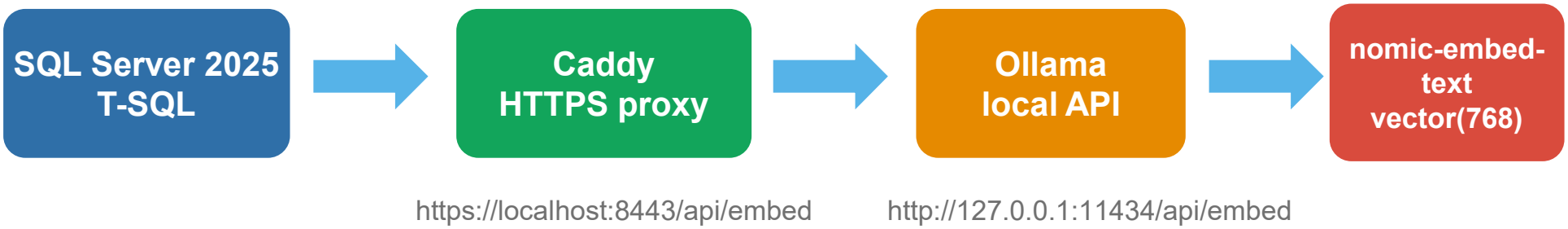


Country-Capital

Key concepts: similarity and efficiency

Ollama	Runs AI models locally, without relying on a cloud provider.
Embedding model	Converts text into embeddings. In this project: nomic-embed-text.
Cosine distance	Compares the orientation of two vectors. Lower distance = greater similarity.
Vector index	Organizes embeddings to speed up similarity search.
DiskANN	Approximate algorithm for finding nearby vector neighbors.

Local project architecture



- SQL Server needs an HTTPS endpoint to communicate with the external service.
- Caddy acts as an HTTPS bridge to Ollama, which runs locally over HTTP.

Complete implementation flow

- The project does not replace the relational model: it adds a semantic layer on top of existing data.
- Structured data + embeddings + SQL queries = integrated semantic search.

SQL Server setup

```
ALTER DATABASE SCOPED CONFIGURATION
SET PREVIEW_FEATURES = ON;

ALTER DATABASE WideWorldImporters
SET COMPATIBILITY_LEVEL = 170;

sp_configure 'external rest endpoint enabled', 1;
RECONFIGURE WITH OVERRIDE;
```

- PREVIEW_FEATURES enables preview vector features.
- Compatibility level 170 aligns the database with SQL Server 2025.
- external rest endpoint allows calls to external services.

- The credential registers the Caddy/Ollama HTTPS endpoint in a controlled way.

Auxiliary table: dbo.ProductSemantic

```
CREATE TABLE dbo.ProductSemantic
(
    ProductSemanticID INT IDENTITY PRIMARY KEY
CLUSTERED,
    StockItemID INT NOT NULL,
    StockItemName NVARCHAR(100) NOT NULL,
    SemanticText NVARCHAR(MAX) NOT NULL,
    Embedding vector(768) NULL
);
```

- The original Warehouse.StockItems table is not modified.
- SemanticText contains the text that will be sent to the model.
- Embedding stores the vector generated for each product.
- StockItemID links each embedding back to its product in Warehouse.StockItems.

Intentional design: separate operational data from the semantic layer.

Registering the external model

```
CREATE EXTERNAL MODEL OllamaNomicEmbed
WITH
(
    LOCATION = 'https://localhost:8443/api/embed',
    API_FORMAT = 'Ollama',
    MODEL_TYPE = EMBEDDINGS,
    MODEL = 'nomic-embed-text'
);
```

- **CREATE EXTERNAL MODEL creates a database object.**
- **It registers where the model is located and how to communicate with it.**
- **It will then be used from T-SQL with USE MODEL OllamaNomicEmbed.**

Generating embeddings inside SQL Server

```
UPDATE dbo.ProductSemantic
SET Embedding =
    CAST(
        AI_GENERATE_EMBEDDINGS(
            SemanticText
            USE MODEL OllamaNomicEmbed
        )
        AS vector(768)
    )
WHERE Embedding IS NULL;
```

- Takes SemanticText from each product.
- Calls Ollama through the external model.
- Obtains a numerical semantic representation.
- CAST AS vector(768) adapts the result to the column type.

Vector index: fast search by meaning

```
CREATE VECTOR INDEX IX_ProductSemantic_Embedding
ON dbo.ProductSemantic (Embedding)
WITH
(
    METRIC = 'cosine',
    TYPE = 'DiskANN'
);
```

- The index is created after all embeddings have been generated.
- cosine measures semantic closeness between text embeddings.
- DiskANN enables approximate nearest-neighbor search.

Without an index: compare against every row.
With an index: locate nearby candidates efficiently.

Semantic search with VECTOR_SEARCH

```
SELECT
    ps.StockItemName AS ProductName
FROM VECTOR_SEARCH
(
    TABLE = dbo.ProductSemantic AS ps,
    COLUMN = Embedding,
    SIMILAR_TO = @queryEmbedding,
    METRIC = 'cosine',
    TOP_N = @top
) AS vs
WHERE CAST(1 - vs.distance AS DECIMAL(19,16)) >= @min_similarity
ORDER BY vs.distance ASC;
```

- **SIMILAR_TO: vector generated from the search text.**
- **METRIC: same metric used by the vector index.**
- **TOP_N: number of nearby candidates.**
- **WHERE: applies a minimum similarity threshold.**
- **ORDER BY: ensures the most relevant results are shown first.**

Three procedures, three levels of output

`find_relevant_product_names_vector_index`	**Simple result**	**Returns only semantically related product names.**
`find_relevant_products_vector_index`	**Technical view**	**Shows SemanticText, distance and similarity.**
`find_relevant_stock_items_vector_index`	**Real-world case**	**Combines semantic search with business data.**

Demo: natural-language queries

```
EXEC dbo.find_relevant_stock_items_vector_index
    @prompt = N'usb gadget',
    @top = 10,
    @min_similarity = 0.3;
```

```
EXEC dbo.find_relevant_stock_items_vector_index
    @prompt = N'funny IT-related gift',
    @top = 10,
    @min_similarity = 0.3;
```

- Expected results: USB novelty drives, IT joke mugs, developer joke mugs.
- The search does not need an exact match: it searches for semantic closeness.
- The @min_similarity threshold helps control result quality.
- Relational filters remain available: color, price, category, supplier.

Best practices and limitations

- The embedding depends on the quality of the text used in SemanticText.
- If the data is in English, English queries usually produce more stable results.
- Vector search can return close results, but not necessarily perfect ones.
- The solution improves when semantic search is combined with traditional SQL filters.
- It does not replace the relational model: it enriches it with a new way to search.

Not exclusive to SQL Server

SQL Server 2025	VECTOR, AI_GENERATE_EMBEDDINGS, VECTOR_SEARCH
Azure SQL Database	vector capabilities in a managed service
PostgreSQL + pgvector	vectors, distances and HNSW/IVFFlat indexes
Oracle Database	integrated AI Vector Search
MySQL HeatWave	vector store, embeddings and semantic search

The trend is clear: bringing AI and vector search into the core of relational databases.

Thank you for your attention

Questions or comments?

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